

Brij Raj Kishore

Staff Software Engineer | C++ | Java | Python

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PROFESSIONAL SUMMARY

Systems software engineer (~6 yrs) in **high-performance C++, Java, and Python** for ML and data systems. Building a **PyTorch out-of-tree device backend for low-latency LLM inference**, on a foundation of query-engine internals (Velox, Gluten, Spark), **JVM** runtimes, and heterogeneous compute (FPGA, CUDA). **18 merged PRs** across PyTorch, vLLM, Meta Velox, Apache Spark, and Apache Gluten; 12 open across 8 repos.

WORK EXPERIENCE

• ZettaBolt Technologies Pvt. Ltd

Noida, India

Senior Member of Technical Staff (Staff Software Engineer)

December 2021 - Present

- **PyTorch OOT Device Backend for LLM Inference**: Building an out-of-tree **C++ device backend** in PyTorch (PrivateUse1) - operator dispatch, kernel registration, device memory, streams - against the **OpenReg** reference device ahead of custom silicon; validated via **OpInfo** and ran **Llama and BERT inference end-to-end**. Built a **CUDA-parity testing framework** + CI. Currently building the **Inductor/Triton codegen path** bringing **torch.compile** to the device, standalone and through **vLLM** serving.
- **Agent Harness for Accelerator-Backend Development**: Built a domain-specific **agent harness** (Claude Code) for PyTorch backend work - **79 files / ~7K lines**: 27 skill areas, **22 subagents**, and **11 hooks** enforcing citation-verified answers, generated-file protection, and blast-radius/test-impact analysis per edit. The **CUDA-parity** and **OpInfo** tooling above run on it.
- **Spark Query Engine Acceleration via FPGA Offloading**: Achieved **5x speedup over vanilla Spark** and **2x over Apache Gluten** by integrating **Velox (C++)** with an FPGA SDK through a **Scala** bridge; designed the operator dispatch architecture for Scan, Filter, and Aggregation.
- **AMD AOCL Integration on Meta Velox**: Delivered **5x runtime improvement** on AMD EPYC by integrating AMD Compiler and AOCL into Meta's Velox/Gluten pipeline (**C++/Scala**); guided by **profiling** with uProf and Flamegraph.
- **Graph Analytics: NVIDIA cuGraph & TigerGraph**: Achieved **100x speedup** on PageRank using TigerGraph GSQL with **NVIDIA CUDA (C++)** and Python; cut pipeline runtime **40%** and infra cost **24%**.
- **ZettaProf - Automated Spark Profiling & Benchmarking Tool**: **Reduced Spark debugging time 60%** by designing and shipping **ZettaProf** (Scala/Java), an automated profiler adopted **company-wide**; mentored engineers on its use.
- **JVM-to-C++ Hive Offload & Healthcare ETL on Microsoft Fabric**: Improved Hive query runtime **20%** migrating execution kernels from the **JVM to a C++ backend** for heterogeneous CPU/FPGA dispatch; separately **cut debugging time ~70%** optimizing healthcare pipelines on **Spark (PySpark/Scala)**, **Microsoft Fabric**, and **Azure Data Factory** (join skew, AQE fallback).

• NXP Semiconductors

Pune, India

Software Engineer

July 2021 - December 2021

Ported BT firmware to dual-antenna platform in **C**; debugged WiFi/BLE driver bugs via GDB; validated via **unit tests**.

• Tata Consultancy Services

Bengaluru, India

Assistant System Engineer

Aug 2017 - Jan 2018

Developed **REST API** endpoints (TCS OmniStore POS); raised unit test coverage **62% to 70%** via JUnit and **CI/CD**.

OPEN SOURCE CONTRIBUTIONS (18 merged, 12 open across 8 upstream repos)

- **Merged**: **PyTorch** (3) - CUDA device-guard RAII fix on the throwing path, LPPoolNd validation, schema-inference errors; **Velox** (5) - TPC-DS map-allocation perf, Spark **transform_values** registration, Parquet encoding; **Gluten** (5) - KeyGroupedPartitioning MatchError fix, codec error messaging, test-suite fixes; **Spark** (4) - config/docs fixes; **vLLM** (1) - Transformers v5 test gating.
- **Under review**: PyTorch fp16 CPU/GPU parity (**clamp, torch.where**); Gluten bloom-filter AQE-fallback corruption; Spark structured streaming (**GroupState**, Protobuf); DuckDB ***COLUMNS()** resolution.

TECHNICAL SKILLS

- **ML / AI**: PyTorch (OOT device backend, PrivateUse1, OpenReg, dispatch, kernel registration, OpInfo, streams), **torch.compile** (Inductor, Triton), CUDA-parity testing, NVIDIA CUDA, cuGraph, vLLM
- **Languages**: C++17, C, **Java**, Scala, Python, Bash
- **Query & Distributed**: Meta Velox, Apache Gluten, Spark SQL, Hive, DuckDB, Parquet/ORC, Hadoop, Kafka, JVM internals, Concurrency
- **Hardware & Tools**: FPGA (Xilinx/AMD), AMD EPYC / AOCL, GDB, Valgrind, uProf, Flamegraph, Docker, Linux, Git, CI/CD
- **Cloud**: AWS, Microsoft Fabric, Azure Data Factory

EDUCATION

• National Institute of Technology (NIT Nagpur)

Nagpur, India

M.Tech - Computer Science & Engineering; CGPA: 8.04 / 10

July 2019 - June 2021

• GL Bajaj Institute of Technology & Management

Greater Noida, India

B.Tech - Computer Science & Engineering

July 2013 - June 2017

ACHIEVEMENTS

Author, [pytorch-internals-explained](#) - public series on PyTorch internals (dispatch, autograd, strides/views, CUDA allocator) | GATE CSE 2019 - **97th Percentile** | Google Code Jam 2020 - **6,448 / 44,434** | TCS CodeVita 2016 - **414 AIR**